

# TIMEMIXER++: A General Time Series Pattern Machine for Universal Predictive Analysis

Shiyu Wang, Jiawei Li, Xiaoming Shi, Zhou Ye, Baichuan Mo, Wenzhe Lin, Shengtong Ju, Zhixuan Chu, Ming Jin

## 01 TIMEMIXER++: Title and Authorship

Authors: Shiyu Wang, Jiawei Li, Xiaoming Shi, Zhou Ye, Baichuan Mo, Wenzhe Lin, Shengtong Ju, Zhixuan Chu, Ming Jin. Affiliations: Griffith University; HKUST Guangzhou; MIT; Zhejiang University; The State Key Laboratory of Blockchain and Data Security; Hangzhou High-Tech Zone Institute of Blockchain and Data Security. Emails: kwuking@gmail.com, jarvis-li@outlook.com, sxm728@hotmail.com, yezhou199032@gmail.com, baichuan@mit.edu, zhixuanchu@zju.edu.cn, {linwenzhe75, shengtongju, mingjinedu}@gmail.com

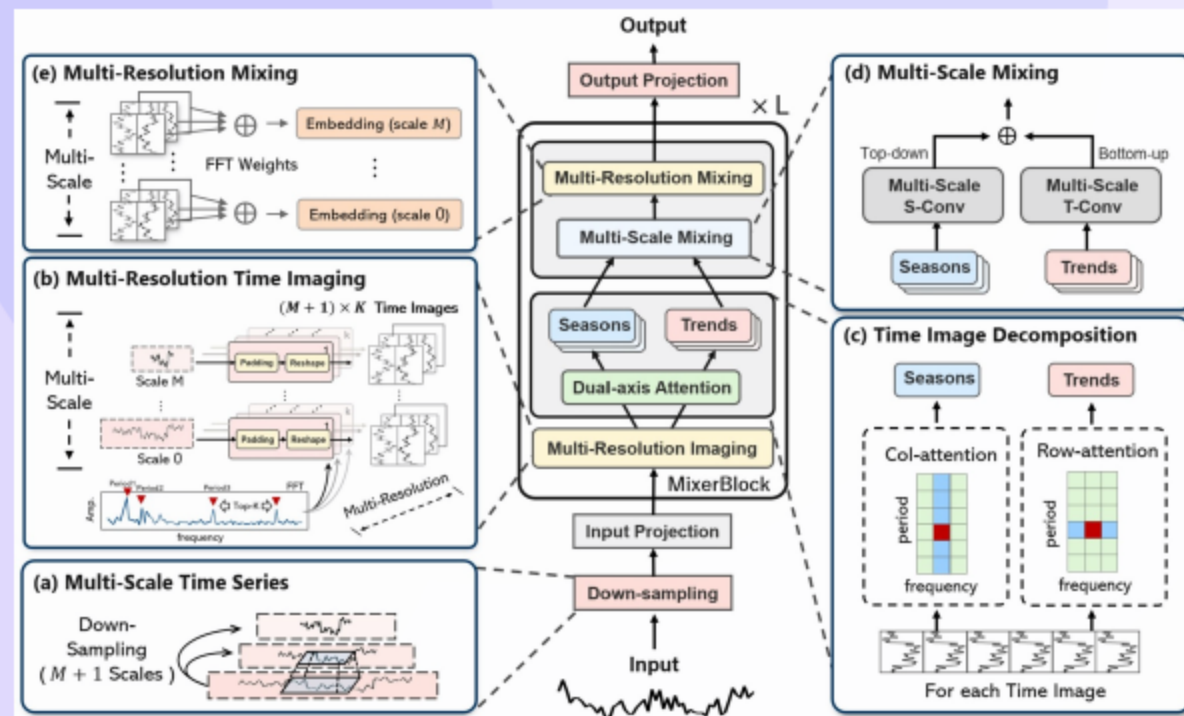
## 02 Motivation: Challenges in Universal TS Modeling

- Universal time series modeling spans forecasting, imputation, classification, and anomaly detection
- Recurrent and convolutional models struggle with long dependencies while attention helps long-range context
- A single robust model must remain effective across tasks, horizons, and scales despite noise and sparsity

## 03 Motivation (cont.): Overlapping Periodicities and Tokenization Gaps

- Overlapping daily, weekly, and seasonal patterns blur token boundaries for sequence models
- Multi-scale and multi-periodicity dynamics demand adaptive pattern extraction without brittle tokenization
- Effective representations should align recurring motifs while tracking evolving trends over time

## 04 Problem Framing: What a General TSPM Must Do



- A general pattern machine captures local motifs and long-range seasonalities without fixed windows
- Frequency-aware adaptation discovers overlapping and time-varying periodicities across contexts
- Cross-variable modeling learns shared structure and idiosyncratic behaviors in multivariate settings
- Task-adaptive representations support forecasting, imputation, classification, and anomaly objectives
- Robustness to sparsity, noise, and shift ensures stable performance under non-stationarity
- Efficiency and interpretability enable scalable learning with clear seasonal and trend views

## 05 Prior Work: Sequence Models for TS (RNN/TCN/Transformer)

- Classical models offer structured seasonality and trend but lack nonlinear flexibility
- Recurrent and convolutional baselines face limits on long horizons and global context
- Transformer and simple MLP variants show **strong long-range performance** in recent studies

## 06 Prior Work: Hierarchical/Decomposition Approaches

- Rigid predefined decompositions limit adaptability to complex and drifting dynamics
- Latent-space disentanglement with dual-axis attention enables flexible seasonal and trend views
- Hierarchical cross-scale interaction improves coverage of multi-periodic structure

## 07 Related SOTA at a Glance

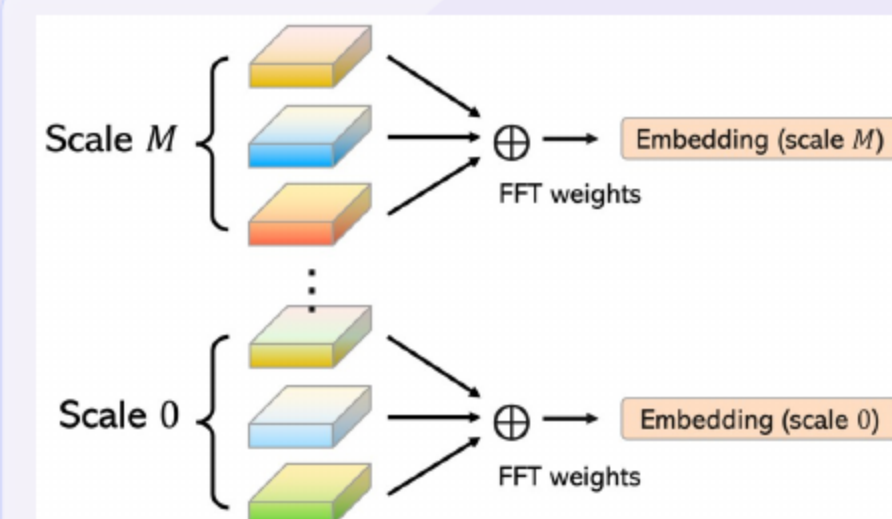
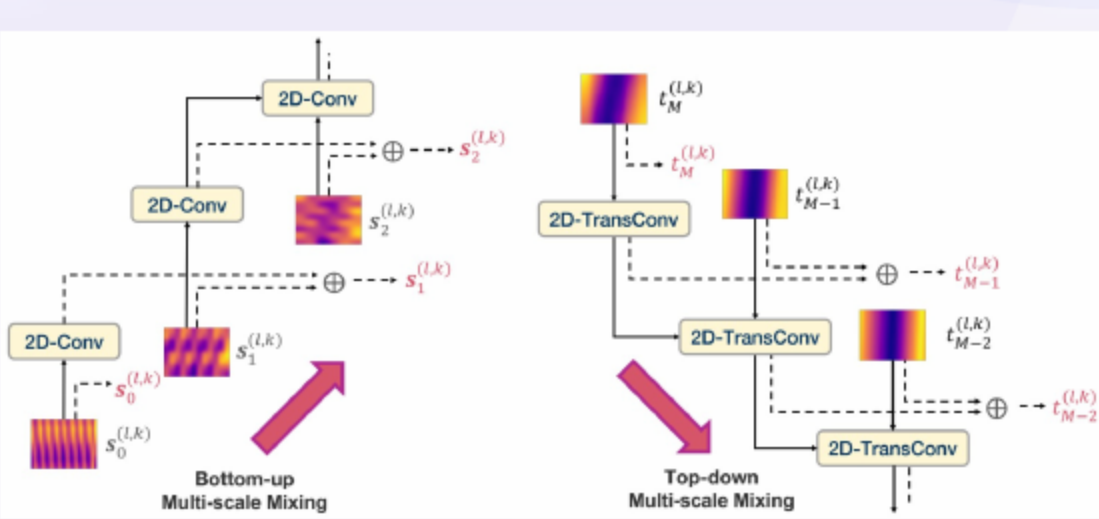
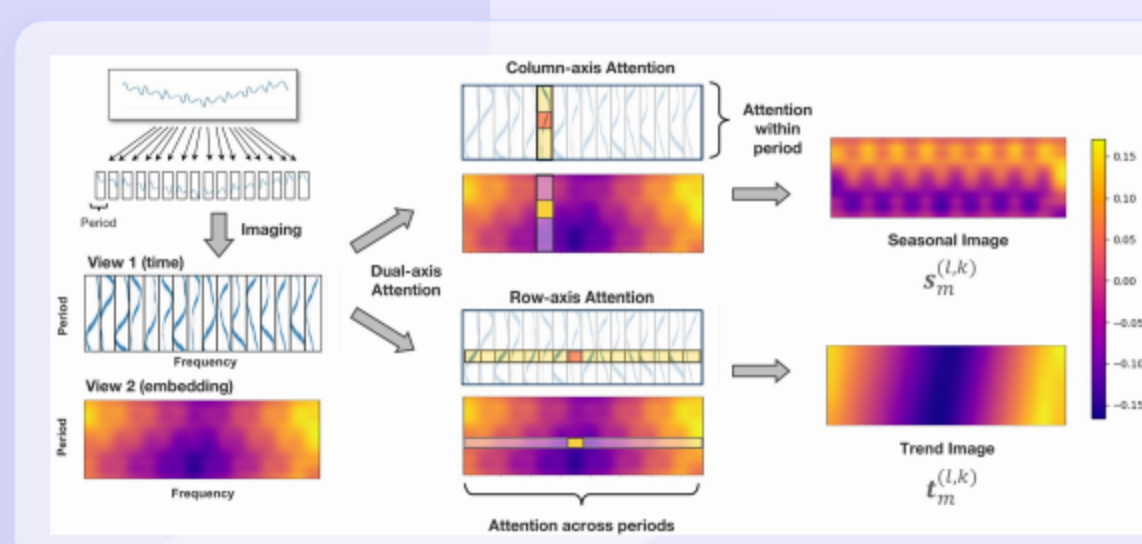
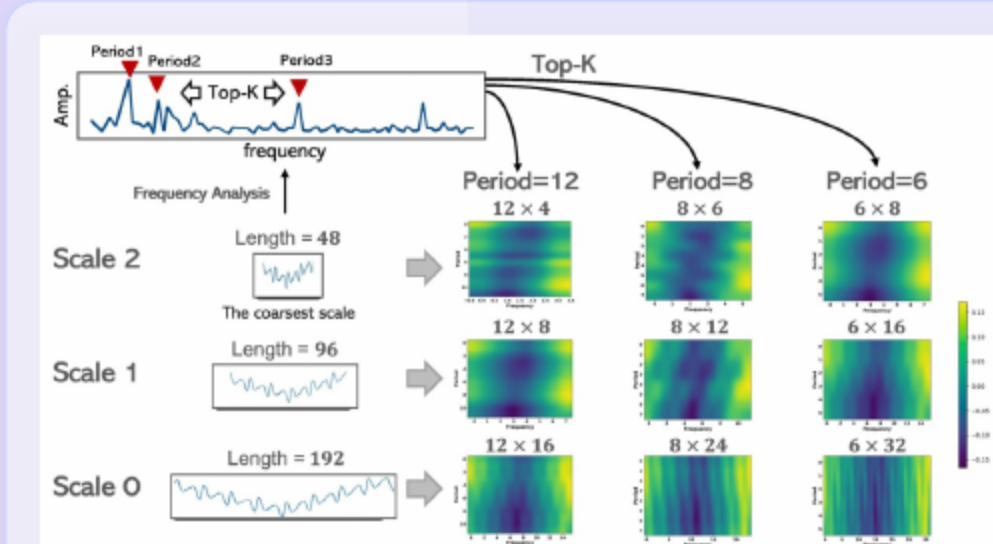
- Strong forecasting baselines include iTransformer, PatchTST, TimesNet, TiDE, and TimeMixer
- Imputation and anomaly tasks often reuse forecasting backbones with masking or detection heads
- Traffic and high-dimensional sensors benchmark multivariate scalability and cross-variable coupling



- Multi-resolution time imaging, dual-axis decomposition, and hierarchical mixing form a unified design
- They report [consistent state-of-the-art](#) across diverse tasks
- They show task-dependent representation profiles that align shared structure with specialized features

- An encoder-only stack projects inputs, processes with MixerBlocks, and aggregates with scale heads
- Channel mixing captures cross-variable dependencies before multi-scale embedding
- Scale-specific predictions are ensembled to match target horizons and tasks

## 10 Inside MixerBlock: MRTI → TID → MCM → MRM



## 11 Ablation Study — why components matter

|                              | ETTh1        | ETTh2        | ETTm1        | ETTm2        | ECL          | Traffic      | Weather      | Solar        | Average      | Promotion |
|------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-----------|
| <b>TIMEMIXER++</b>           | <b>0.419</b> | <b>0.339</b> | <b>0.369</b> | <b>0.269</b> | <b>0.165</b> | <b>0.416</b> | <b>0.226</b> | <b>0.203</b> | <b>0.300</b> | <b>-</b>  |
| w/o channel mixing           | 0.424        | 0.346        | 0.374        | 0.271        | 0.197        | 0.442        | 0.233        | 0.245        | 0.317        | 5.36%     |
| w/o time image decomposition | 0.441        | 0.358        | 0.409        | 0.291        | 0.198        | 0.445        | 0.251        | 0.241        | 0.329        | 8.81%     |
| w/o multi-scale mixing       | 0.447        | 0.361        | 0.391        | 0.284        | 0.172        | 0.427        | 0.239        | 0.234        | 0.320        | 6.25%     |
| w/o multi-resolution mixing  | 0.431        | 0.350        | 0.374        | 0.280        | 0.181        | 0.432        | 0.241        | 0.233        | 0.316        | 5.10%     |

- Removing time image decomposition, they observe **large degradation** on overlapping and drifting periodicities
- Eliminating multi-scale mixing, they observe weakened long-horizon accuracy due to reduced cross-scale flow
- Disabling multi-resolution fusion and channel mixing, they report reduced stability and multivariate gains

12

Representation Analysis — what the model learns

- Seasonal and trend paths remain distinct while aligning motifs and tracking baselines
- Representation similarity profiles vary by task to balance shared and specialized features
- Task adaptive structure supports better generalization across objectives and domains

13

Results Across 8 Tasks (brief setup)

| Models       | TimeMixer++<br>(Ours) |       | TimeMixer<br>(2024b) |       | iTransformer<br>(2024) |       | PatchTST<br>(2023) |       | Crossformer<br>(2023) |       | TiDE<br>(2023a) |       | TimesNet<br>(2023) |       | DLinear<br>(2023) |       | SCINet<br>(2022a) |       | FEDformer<br>(2022b) |       | Stationary<br>(2022c) |       | Autoformer<br>(2021) |       |
|--------------|-----------------------|-------|----------------------|-------|------------------------|-------|--------------------|-------|-----------------------|-------|-----------------|-------|--------------------|-------|-------------------|-------|-------------------|-------|----------------------|-------|-----------------------|-------|----------------------|-------|
|              | MSE                   | MAE   | MSE                  | MAE   | MSE                    | MAE   | MSE                | MAE   | MSE                   | MAE   | MSE             | MAE   | MSE                | MAE   | MSE               | MAE   | MSE               | MAE   | MSE                  | MAE   | MSE                   | MAE   | MSE                  | MAE   |
| Electricity  | 0.165                 | 0.253 | 0.182                | 0.272 | 0.178                  | 0.270 | 0.205              | 0.290 | 0.244                 | 0.334 | 0.251           | 0.344 | 0.192              | 0.295 | 0.212             | 0.300 | 0.268             | 0.365 | 0.214                | 0.327 | 0.193                 | 0.296 | 0.227                | 0.338 |
| ETT (Avg)    | 0.349                 | 0.399 | 0.367                | 0.388 | 0.383                  | 0.377 | 0.381              | 0.397 | 0.685                 | 0.578 | 0.482           | 0.470 | 0.391              | 0.404 | 0.442             | 0.444 | 0.689             | 0.597 | 0.408                | 0.428 | 0.471                 | 0.464 | 0.465                | 0.459 |
| Exchange     | 0.357                 | 0.391 | 0.391                | 0.453 | 0.378                  | 0.360 | 0.403              | 0.404 | 0.940                 | 0.707 | 0.370           | 0.413 | 0.416              | 0.443 | 0.354             | 0.414 | 0.750             | 0.626 | 0.519                | 0.429 | 0.461                 | 0.454 | 0.613                | 0.539 |
| Traffic      | 0.416                 | 0.264 | 0.484                | 0.297 | 0.428                  | 0.282 | 0.481              | 0.304 | 0.550                 | 0.304 | 0.760           | 0.473 | 0.620              | 0.336 | 0.625             | 0.383 | 0.804             | 0.509 | 0.610                | 0.376 | 0.624                 | 0.340 | 0.628                | 0.379 |
| Weather      | 0.226                 | 0.262 | 0.240                | 0.271 | 0.258                  | 0.278 | 0.259              | 0.281 | 0.259                 | 0.315 | 0.271           | 0.320 | 0.259              | 0.287 | 0.265             | 0.317 | 0.292             | 0.363 | 0.309                | 0.360 | 0.288                 | 0.314 | 0.338                | 0.382 |
| Solar-Energy | 0.203                 | 0.238 | 0.216                | 0.280 | 0.233                  | 0.262 | 0.270              | 0.307 | 0.641                 | 0.639 | 0.347           | 0.417 | 0.301              | 0.319 | 0.330             | 0.401 | 0.282             | 0.375 | 0.291                | 0.381 | 0.261                 | 0.381 | 0.885                | 0.711 |

- They evaluate across 8 tasks and 30 benchmarks
- They observe consistent improvements over 27 baselines

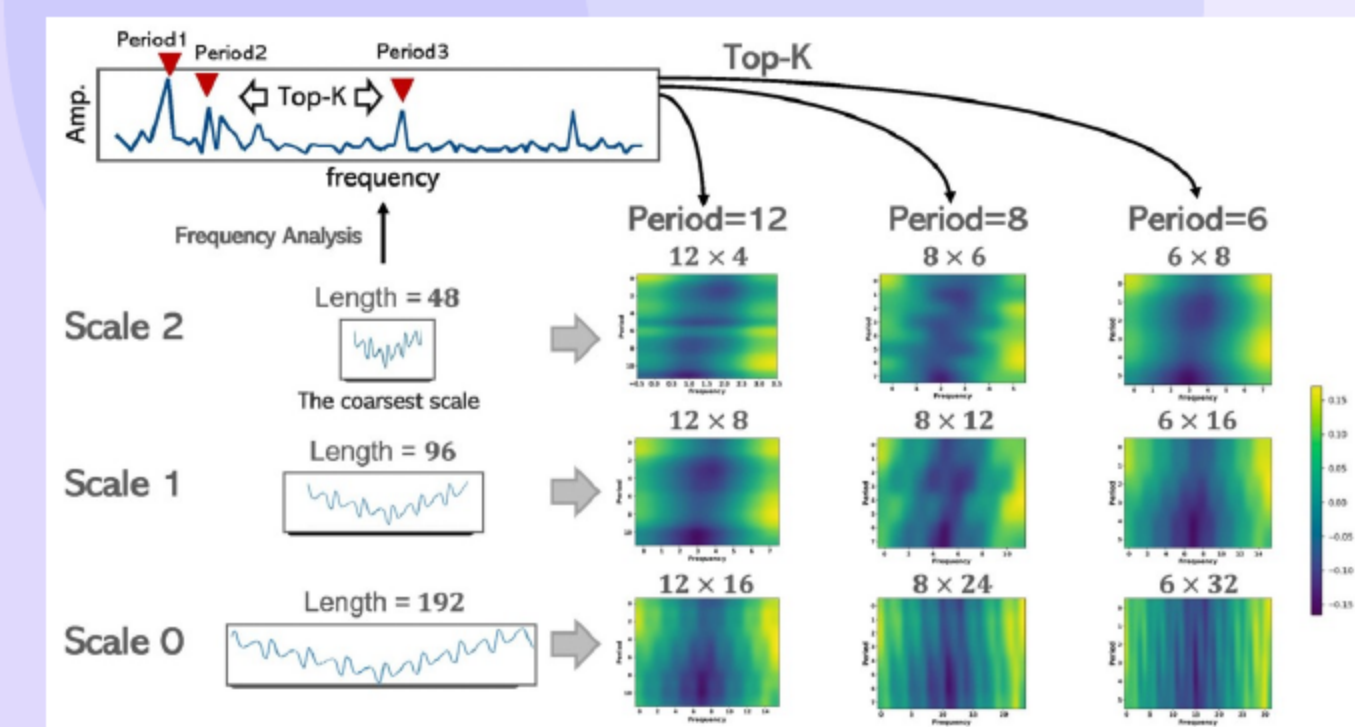
14

From Motivation to Practice: When and Why TIMEMIXER++ Helps

- These settings benefit; they show that time images and dual-axis decomposition help with multi-periodic, cross-scale dynamics
- These settings benefit; they show gains on high-dimensional sensors from cross-variable channel mixing and scale ensembles
- These settings benefit; they show improvements in sparse data and transfer with adjustable frequency count and capacity placement

15

Main Failure Mode: Sensitivity to Frequency Selection and Weak Periodicity



- Top period selection is sensitive to noisy spectra and mis specified counts during imaging
  - Mitigate with learned frequency gating and broader candidate pools
  - Stabilize with spectral jitter and periodic re estimation
- Weak or non stationary seasonality undermines frequency derived representations
  - Fallback to time domain paths when confidence is low
  - Use change point cues to damp frequency contributions



- The approach is **strong overall** yet remains sensitive to frequency selection in challenging regimes
- They propose future work including adaptive frequency discovery and richer time-domain fallbacks
- They plan broader benchmarks and deployment studies to validate scalability and robustness

- 
- They deliver clear seasonal and trend decomposition via time images and dual-axis attention
  - They produce **robust performance** across tasks through cross-scale and cross-resolution mixing
  - They support efficient and transferable modeling with a unified encoder and scale heads
- 

# Thank You

Questions & Discussion